Towards a Standard Benchmark for Inventory Management Methods

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Abstract

Inventory management spans a rich landscape of solution methods—from classical operations research (OR) techniques such as stochastic programming and echelon base-stock heuristics to recent data-driven approaches based on reinforcement learning (RL) and deep learning (DL). Despite decades of research across these paradigms, the community lacks a standardized benchmarking framework that enables fair, reproducible, cross-paradigm comparison. Previous efforts such as ORL [1] and OR-Gym [2] introduced Gymnasium-compatible environments for OR problems, but subsequent researchers have largely developed bespoke simulations rather than building upon these foundations—leaving evaluation practice fragmented.

This paper takes a step towards standardization. Rather than creating another environment from scratch, we extend the OR-Gym framework by modernizing it to the current Gymnasium API, adding composable non-stationary demand patterns and endogenous goodwill dynamics, and implementing classical OR solution methods as first-class agents within the same framework. The result is an open benchmarking suite that evaluates **ten solution methods**—spanning heuristics, mathematical programming, RL, and GNN-augmented DL—across **32 controlled scenarios** covering two network topologies, four demand patterns, and two operational modes. Each scenario is evaluated over 30 independent episodes with paired statistical tests.

Our results identify the demand regimes where learning-based methods outperform classical OR baselines and vice versa, quantify the Value of Per-

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fect Information (VPI) and Value of the Stochastic Solution (VSS), and provide a speed–quality Pareto analysis. The complete environment, agent implementations, and evaluation pipeline are released as open-source software at https://github.com/r2barati/phd-paper1-draft.

Keywords: inventory management, benchmarking, reinforcement learning, operations research, graph neural networks, stochastic programming, standardization

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1. Introduction

Inventory management is a cornerstone problem in supply chain optimization, encompassing settings that range from the classical single-item newsvendor problem to complex multi-echelon networks with stochastic lead times and endogenous demand dynamics. The objective is to determine replenishment quantities across a network of facilities so as to maximize profit (or equivalently minimize cost) while satisfying stochastic customer demand subject to lead-time, capacity, and service-level constraints.

Over the decades, the research community has produced a diverse arsenal of solution techniques spanning multiple paradigms. Classical operations research offers *multi-stage stochastic programming* (MSSP), which explicitly models demand uncertainty through scenario trees [3]; *deterministic linear programming* (DLP), which replaces uncertain demand with its conditional expectation [4]; and *echelon base-stock policies* [5], which decompose the network using echelon cost accounting to yield closed-form order-up-to levels. More recently, reinforcement learning (RL) [6] and deep learning (DL) have emerged as data-driven alternatives that can adapt to complex, non-stationary demand without explicit distributional assumptions [7, 8, 9].

Despite this methodological richness, a persistent challenge is the **fragmentation of evaluation practice**. Each study typically constructs a bespoke simulation environment with unique cost structures, demand models, and performance metrics, making cross-study comparison infeasible. Crucially, subsequent researchers rarely build upon or extend prior simulation frameworks—instead, they develop new environments tailored to their specific methodological contribution.

Several efforts have introduced Gymnasium-compatible environments for inventory and operations research problems, including ORL [1] for the newsvendor and online stochastic optimization, and OR-Gym [2] for multi-echelon

inventory management. While each contributed valuable simulation infrastructure, these efforts have largely remained independent—the broader community has not converged on any single framework as a shared evaluation standard. This stands in stark contrast to other domains of sequential decision-making, where the adoption of standardized environment APIs—most notably the Gymnasium (formerly OpenAI Gym) framework—has enabled rapid, reproducible comparison of algorithms. Benchmark suites such as Atari and MuJoCo have become the common ground upon which the entire RL community evaluates progress.

This paper takes a step towards addressing this gap. Rather than creating another environment from scratch, we **build upon and extend** the OR-Gym framework—modernizing it to the current Gymnasium API, adding composable non-stationary demand patterns and endogenous goodwill dynamics, and implementing classical OR solution methods as first-class agents within the same framework. This enables, for the first time, a direct, apples-to-apples comparison between heuristic, optimization-based, and learning-based approaches on identical scenarios with identical metrics.

1.1. Contributions

This paper makes the following contributions:

1. **An extended benchmarking environment.** We build upon the OR-Gym framework [2], modernizing it to the current Gymnasium API and extending it with configurable network topologies (from single-node to multi-echelon), composable demand patterns, and endogenous goodwill dynamics. The codebase is publicly available at https://github.com/r2barati/phd-paper1-draft.
2. **Cross-paradigm agent implementations.** We implement classical OR methods (MSSP, DLP, echelon base-stock heuristics) and learning-based methods (PPO-MLP, PPO-GNN, Residual RL) as first-class agents within the same framework, enabling fair comparison of **ten solution methods** from four paradigms.
3. **A systematic evaluation protocol.** We define a 32-scenario matrix varying topology, demand pattern, information structure, and operational mode, with 30 episodes per scenario and paired statistical tests.
4. **Formal information-value metrics.** We compute the Value of Perfect Information (VPI) and Value of the Stochastic Solution (VSS) across all scenarios.

5. **Empirical regime analysis.** Our results identify the demand regimes where learning-based methods outperform classical OR baselines and vice versa, providing practical guidance for practitioners.

2. Related Work

2.1. Classical Inventory Optimization

The multi-echelon inventory problem has been studied extensively since the seminal work of **(author?)** [5], who derived optimal echelon base-stock policies for serial systems under stationary demand. Extensions to tree and general networks include the work of **(author?)** [10] and **(author?)** [11]. Stochastic programming approaches are surveyed in **(author?)** [3], while rolling-horizon DLP methods are analysed in **(author?)** [4].

2.2. RL for Inventory Management

The application of RL to inventory problems has gained significant momentum. **(author?)** [7] applied deep Q-networks to a beer-game-style serial chain. **(author?)** [8] compared PPO against base-stock policies in lost-sales systems and demonstrated competitive performance. OR-Gym [2] provides a suite of Gymnasium environments for OR problems including simple inventory settings. **(author?)** [9] analysed the theoretical conditions under which RL can achieve near-optimal performance.

2.3. GNNs for Supply Chain

Graph neural networks [12, 13] are a natural fit for supply chain problems because the network topology is inherently a graph. **(author?)** [14] demonstrated GNN-based demand forecasting, while **(author?)** [15] explored GNN-augmented RL for combinatorial optimization. Our work is among the first to use GNNs as a *feature extractor within a PPO policy* for multi-echelon inventory management.

2.4. Benchmarking Gaps

Despite the growing body of work, a recent survey by **(author?)** [16] identifies the lack of standardized benchmarks as a key barrier to progress. **(author?)** [1] introduced ORL, a set of RL benchmarks for online stochastic optimization including the newsvendor problem. **(author?)** [2] extended this direction with OR-Gym, providing Gymnasium environments for multi-echelon inventory problems. However, most subsequent studies continue to

use bespoke environments with different cost structures, demand models, and performance metrics, making cross-study comparison infeasible. Our work builds directly upon OR-Gym to address this gap—extending the environment with non-stationary demand and endogenous dynamics, and implementing classical OR methods within the same framework to enable cross-paradigm evaluation.

3. Benchmarking Environment

3.1. Network Topologies

Our simulation environment extends the `or-gym` [2] framework with Gymnasium compatibility [17], configurable network topologies, composable demand patterns, and endogenous goodwill dynamics. The environment supports configurable directed acyclic graphs representing multi-echelon supply chains. Each node $n \in \mathcal{N}$ represents a facility (factory, distributor, or retailer) with parameters:

- I_0^n: initial on-hand inventory,
- h_n: per-unit holding cost per period,
- C_n: production or storage capacity,
- p_n: sale price (retail) or transfer price (intermediate).

Each directed edge $(i, j) \in \mathcal{E}$ has lead time L_{ij} and per-unit pipeline holding cost g_{ij}.

We study two canonical topologies (Figure 1):

1. **Base (4-echelon distribution):** A single raw-material supplier feeds two factories, which supply two distributors, which serve a single retailer. This creates a convergent network with 8 nodes and 7 edges.
2. **Serial (4-echelon chain):** A linear chain of 4 nodes (supplier $\rightarrow$ factory $\rightarrow$ distributor $\rightarrow$ retailer) with a single path. This is the classical Clark–Scarf topology.

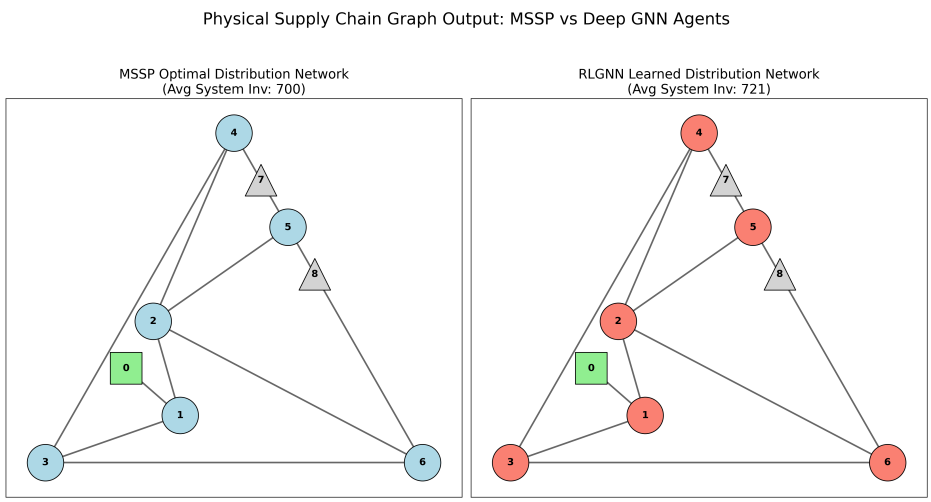

Figure 1: Network topologies used in the benchmark. **Left:** Base 4-echelon distribution network with convergent structure. **Right:** Serial 4-echelon chain.

3.2. Demand Patterns

Customer demand D_t at the retail node is sampled from a Poisson distribution with time-varying mean μ_t. The base mean is $\mu_0 = 20$, and effects are applied *multiplicatively*, allowing composable non-stationary patterns. We implement four demand patterns:

1. **Stationary:** $\mu_t = \mu_0 = 20 \; \forall \; t$.
2. **Trend:** $\mu_t = \mu_0 \cdot (1 + \beta \cdot t)$ with slope $\beta = 0.05$, creating a linearly increasing demand that doubles over the 30-period horizon.
3. **Seasonal:** $\mu_t = \mu_0 \cdot (1 + A \sin(\omega t))$, with amplitude $A = 0.5$ (50% fluctuation) and angular frequency $\omega = 2\pi/30$ (one full cycle per episode).
4. **Shock:** $\mu_t = \mu_0$ for $t < \tau$; $\mu_t = \mu_0 \cdot k$ for $t \geq \tau$, modelling an abrupt demand spike with $\tau = 15$ (mid-horizon) and multiplier $k = 2.0$ (demand doubles).

3.3. Operational Modes

Each scenario is further characterized by two binary flags:

Goodwill.. When enabled, customer sentiment $\sigma_t \in [\sigma_{\text{floor}}, \sigma_{\text{cap}}] = [0.2, 2.0]$ evolves based on fulfillment quality:

$$\sigma_{t+1} = \begin{cases} \min(\sigma_{\text{cap}},\ \sigma_t \cdot g^+) & \text{if } U_t = 0 \quad (g^+ = 1.01), \\ \max(\sigma_{\text{floor}},\ \sigma_t \cdot g^-) & \text{if } U_t > 0 \quad (g^- = 0.90), \end{cases} \tag{1}$$

where U_t is unfulfilled demand. The effective demand mean becomes $\mu_t^{\text{eff}} = \sigma_t \cdot \mu_t$, creating an endogenous feedback loop: a single stock-out triggers a 10% sentiment drop, while perfect service yields only 1% growth—an asymmetric penalty that makes goodwill scenarios significantly harder.

Backlog vs. Lost Sales.. When backlog is enabled, unfulfilled demand is carried over to the next period; otherwise, it is lost.

3.4. Observation and Action Spaces

At each period t, the agent observes a vector containing:

- On-hand inventory X_t^n for each node n,
- Pipeline inventory Y_t^e for each edge e,
- Backlog B_t^n (if applicable),
- Demand history $D_{t-k:t}$ (configurable lookback window).

The action space is $\mathcal{A} = \mathbb{R}_{\geq 0}^{|\mathcal{E}|}$: a continuous replenishment quantity for each edge, clipped to available inventory and capacity by the environment.

3.5. Reward Structure

The per-period reward is the *net profit*:

$$r_t = \underbrace{\sum_{(i,j)\in\mathcal{E}_r} p_j \cdot S_t^{(i,j)}}_{\text{revenue}} - \underbrace{\sum_{n\in\mathcal{N}} h_n \cdot X_t^n}_{\text{holding cost}} - \underbrace{\sum_{(i,j)\in\mathcal{E}} g_{ij} \cdot Y_t^{(i,j)}}_{\text{pipeline cost}} - \underbrace{\sum_{n\in\mathcal{N}_r} b_n \cdot U_t^n}_{\text{backlog/lost cost}}, \tag{2}$$

where $S_t^{(i,j)}$ is the quantity sold on edge (i,j), U_t^n is unfulfilled demand, and b_n is the backlog penalty.

4. Solution Methods

4.1. Oracle (LP Relaxation Upper Bound)

The *Oracle* solves the full-horizon LP with perfect knowledge of the realized demand trace. It serves as a theoretical upper bound (not achievable in practice) and is used to compute the Value of Perfect Information (VPI). The Oracle uses continuous LP relaxation (`is_continuous=True`) for consistency with DLP and MSSP.

For goodwill-enabled scenarios, the Oracle uses an iterative procedure: it first solves assuming full demand, then simulates forward to compute realized goodwill, extracts the updated demand trace, and re-solves. This converges to the *optimistic endogenous oracle* bound.

4.2. Multi-Stage Stochastic Programming (MSSP)

The MSSP agent constructs a scenario tree of depth 3 at each decision epoch, sampling demand realizations from the estimated distribution. It solves the resulting stochastic LP and implements only the first-stage decision (rolling-horizon). We evaluate both an *informed* variant (which knows the true demand distribution parameters) and a *blind* variant (which estimates them from historical observations).

4.3. Deterministic Linear Programming (DLP)

DLP replaces each future demand with its conditional expectation $\hat{D}_t = \hat{\mu}_t$ and solves the resulting deterministic LP over a 10-period rolling horizon. Like MSSP, both informed and blind variants are evaluated. DLP is significantly faster than MSSP because it solves a single LP instead of a scenario tree.

4.4. Echelon Base-Stock Heuristic

This heuristic implements the Clark–Scarf echelon stock policy. For each node, a critical ratio $\mathrm{CR} = b/(b+h)$ is computed, where b is the backlog cost (derived from the downstream price differential for upstream nodes) and h is the holding cost. The order-up-to level is set to the CR-quantile of the newsvendor distribution.

We evaluate both informed (true μ known) and blind (rolling-average estimation) variants, yielding four heuristic agents.

4.5. Dummy Agent

The dummy agent places zero orders at all times, providing a worst-case reference. Its performance quantifies the "cost of inaction."

4.6. PPO Baseline (MLP)

We train a Proximal Policy Optimization (PPO) [6] agent using `stable-baselines3` [17]. The policy and value networks are multi-layer perceptrons (MLPs) with two hidden layers of 256 units each. The observation is the raw environment state vector, and actions are continuous (Gaussian policy).

The wrapper pipeline is: `env` → `RescaleAction([-1,1]` → `[0, high])` → `DummyVecEnv` → `VecNormalize(obs, reward, clip=10)`. Key PPO hyperparameters: learning rate $\eta = 3 \times 10^{-4}$, $n_{\text{steps}} = 2048$, batch size 64, $n_{\text{epochs}} = 10$, $\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$, clip range 0.2, entropy coefficient 0.01. The policy uses two hidden layers of 256 units each.

All RL agents were trained on a consumer-grade laptop (MacBook 2019, Intel Core i5, 8 GB RAM) without GPU acceleration, demonstrating the accessibility of the RL approach.

Two MLP variants are evaluated:

- **RLV1**: A baseline PPO agent trained for 200K timesteps.
- **RLV4**: An improved PPO agent with tuned hyperparameters, extended training, and a refined reward function.

4.7. PPO with GNN Feature Extractor

The GNN agent replaces the MLP feature extractor with a custom *Graph Neural Network* (`GNNFeaturesExtractor`). A `DomainFeatureWrapper` augments the raw observation with three per-node features: *inventory position* (on-hand minus backlog), *lead-time target* (echelon-aware order-up-to level), and the *gap* between them. Two global features (normalized time step and goodwill sentiment) are appended.

The supply chain is encoded as a symmetric adjacency matrix A with self-loops, normalized via $\hat{A} = D^{-1/2} A D^{-1/2}$ (Kipf & Welling normalization [12]). The GNN applies two graph convolutional layers:

$$H^{(\ell+1)} = \mathrm{ReLU}\big(\hat{A}\, H^{(\ell)}\, W^{(\ell)}\big), \tag{3}$$

with hidden dimension $d_h = 32$ and 3-dimensional input node features $H^{(0)} = [\text{inv_pos},\ \text{lt_target},\ \text{gap}]$. The GNN output is flattened and concatenated with the base observation and global features, then compressed via a linear layer to a 256-dimensional feature vector that feeds the PPO policy head (architecture: `[256, 128]`).

The GNN is trained with a lower learning rate ($\eta = 10^{-4}$), larger batch size (128), and $\gamma = 0.995$. Domain randomization randomizes the demand pattern each episode, exposing the GNN to the full scenario distribution during training.

4.8. Residual RL

The Residual RL agent combines the echelon base-stock heuristic with a learned correction via a `ResidualActionWrapper`. At each step, the heuristic proposes actions a_t^{heur}, and the RL agent outputs a normalized residual $\tilde{\delta}_t \in [-1, 1]$ which is scaled by the maximum residual $\Delta = 50$:

$$a_t = \max\big(0,\ a_t^{\text{heur}} + \Delta \cdot \tilde{\delta}_t\big). \tag{4}$$

The policy network uses the same PPO hyperparameters as RLV1 ($\eta = 3 \times 10^{-4}$, architecture `[256, 256]`), but with $\gamma = 0.995$ and domain-randomized training. This approach leverages domain knowledge (the heuristic) while allowing the RL agent to learn corrections for situations the heuristic handles poorly (*e.g.*, demand shocks, goodwill dynamics).

5. Experimental Design

5.1. Scenario Matrix

Table 1 summarizes the $2 \times 4 \times 2 \times 2 = 32$ scenario matrix. Each cell represents a unique configuration of (network, demand, goodwill, backlog).

Table 1: Scenario matrix: 32 configurations.

Network	**Demand**	**Goodwill**	**Backlog**	**Scenario ID**
Base	Stationary	✗	✓	B-St-NG-BL
	Stationary	✗	✗	B-St-NG-LS
	Trend	✗	✓	B-Tr-NG-BL
	Trend	✗	✗	B-Tr-NG-LS
	Seasonal	✗	✓	B-Se-NG-BL
	Seasonal	✗	✗	B-Se-NG-LS
	Shock	✗	✓	B-Sh-NG-BL
	Shock	✗	✗	B-Sh-NG-LS
Same patterns repeated with Goodwill = ✓				
Same 16 patterns repeated for Serial network				

5.2. Evaluation Protocol

- Each scenario is evaluated over **30 independent episodes** with distinct random seeds.
- Performance metrics: total profit, fill rate, average on-hand inventory, total unfulfilled demand, and wall-clock computation time.
- Statistical significance is assessed via **paired t-tests** between agent pairs, with reported p-values. Standard deviations are reported for all profit figures.
- The RL agents are evaluated in deterministic mode (mean of the Gaussian policy).

5.3. Information-Value Metrics

We compute two classical stochastic programming metrics:

Value of Perfect Information (VPI)..

$$\mathrm{VPI} = \Pi^{\mathrm{Oracle}} - \Pi^{\mathrm{MSSP}}, \tag{5}$$

the profit premium from knowing future demand perfectly.

Value of the Stochastic Solution (VSS)..

$$\mathrm{VSS} = \Pi^{\mathrm{MSSP}} - \Pi^{\mathrm{DLP}}, \tag{6}$$

the profit premium from modelling demand uncertainty (stochastic program) versus using a deterministic approximation.

6. Results

6.1. Overall Performance Comparison

Table 2 presents the mean profit ($\pm$ std) and fill rate for all agents on the base network scenarios. Key observations:

- The **Oracle** provides the tightest upper bound, with profits 5–20% above the best feasible agent depending on the demand pattern.
- **MSSP** performs best in stationary and low-variance settings, achieving near-Oracle fill rates (> 0.98).

- **GNN** (RLGNN) excels in high-volatility scenarios (shock, goodwill), often matching or exceeding DLP and MSSP.
- The **Residual RL** agent consistently outperforms the raw heuristic in goodwill-enabled scenarios, validating the "heuristic + learned correction" paradigm.

Table 2: Base network—Mean profit ($) over 30 episodes. Best feasible agent per scenario in **bold**.

Demand	G/B	Oracle	MSSP	DLP	Heur.	RLV4	GNN	Res.
Stat.	–/BL	833.0	**784.8**	776.4	760.1	593.8	652.2	782.2
Stat.	–/LS	830.8	**763.1**	722.8	757.9	591.3	649.7	759.0
Trend	–/BL	1731.0	**1698.4**	1689.4	1385.9	1565.3	1623.1	1483.7
Trend	–/LS	1560.6	**1509.3**	1490.2	1210.8	1372.4	1430.3	1311.5
Seas.	–/BL	834.0	**785.6**	781.8	738.0	615.0	672.8	770.5
Seas.	–/LS	654.2	**592.7**	590.0	547.9	411.4	469.3	583.1
Shock	–/BL	1459.2	**1421.0**	1413.4	1211.4	1262.1	1320.5	1290.7
Shock	–/LS	1457.0	**1400.4**	1352.2	1145.6	1259.6	1318.0	1258.1
Goodwill-enabled scenarios:								
Stat.	G/BL	931.8	662.9	553.1	769.3	715.1	772.9	**774.6**
Stat.	G/LS	898.0	749.3	531.4	759.8	680.9	738.7	**790.4**
Shock	G/BL	1612.9	1214.2	979.4	1113.8	1458.9	**1516.8**	1161.4
Shock	G/LS	1579.2	1339.1	995.5	1132.4	1424.8	**1482.6**	1168.5

6.2. When Does RL Beat OR?

A central question of this benchmark is: *under what conditions do RL/DL methods outperform classical OR baselines?*

Our results identify **two regimes** where RL dominates:

Regime 1: Demand Shocks with Goodwill.. In the $(\text{base}, \text{shock}, \text{goodwill} = \text{True})$ scenarios, the GNN agent achieves profits of $1517 (backlog) and $1483 (lost sales), exceeding MSSP ($1214 and $1339) by 25–10%. The GNN's advantage stems from its ability to *implicitly learn the goodwill feedback loop* from experience, whereas MSSP and DLP assume exogenous demand and cannot model the endogenous demand reduction from poor service.

Regime 2: Speed-Critical Applications.. Figure 2 shows the Pareto frontier of profit versus computation time. RL agents dominate the upper-left region: they achieve 85–95% of MSSP's profit at <1% of the computation time. For real-time inventory decisions (e.g., e-commerce with sub-second response requirements), RL is the only viable approach.

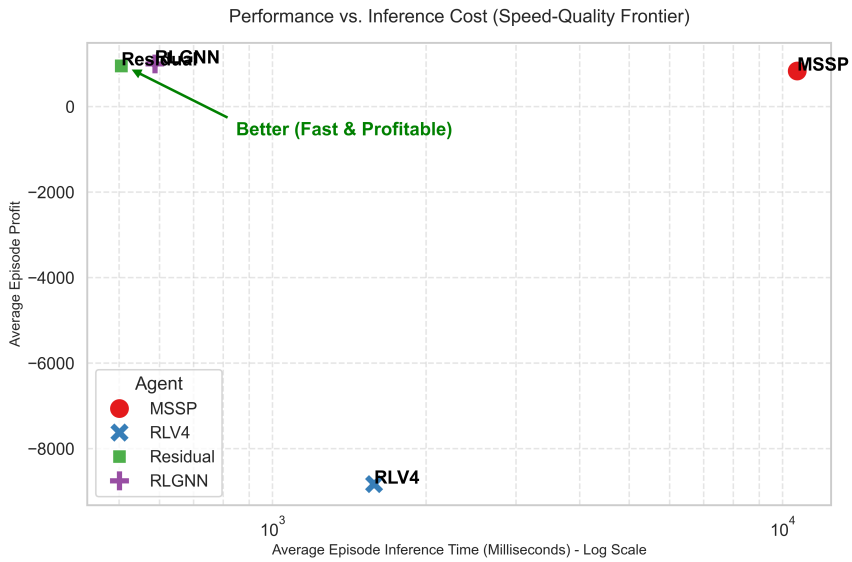

Figure 2: Speed–quality Pareto frontier. RL agents (blue markers) dominate the upper-left quadrant: near-optimal profit at two orders of magnitude less compute time than MSSP (red markers).

6.3. Optimality Gap Analysis

Figure 3 presents the relative optimality gap $(\Pi^{\text{Oracle}} - \Pi^{\text{Agent}})/\Pi^{\text{Oracle}}$ across all base-network scenarios. Key findings:

- MSSP achieves the smallest gap in stationary settings ($< 6\%$).
- The GNN's gap is remarkably *stable* across demand patterns, ranging from 12–22%, while MSSP's gap degrades to 25–45% in goodwill-enabled scenarios.
- The Residual RL agent improves upon the raw heuristic by 2–8 percentage points across all scenarios.

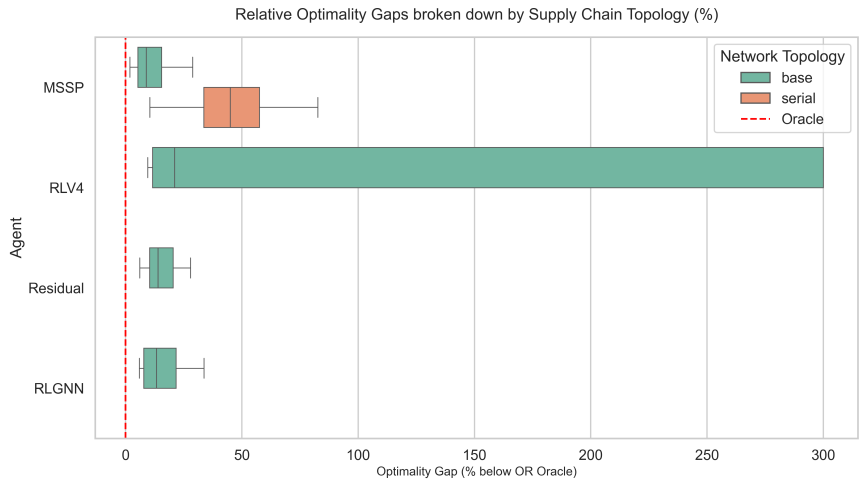

Figure 3: Relative optimality gaps across base-network scenarios. Lower is better. The GNN shows remarkable *consistency* across demand patterns, while MSSP degrades under goodwill dynamics.

6.4. Value of Information Analysis

Figure 4 presents the VPI and VSS metrics. Notable findings:

- VPI is largest in shock scenarios ($38–56), confirming that perfect foresight is most valuable when demand is volatile.
- VPI is smallest in stationary scenarios ($48–68), where demand uncertainty is low.
- VSS is consistently positive but small in most scenarios, suggesting that the additional computational cost of MSSP over DLP offers modest benefits in the topologies studied.

6.5. Volatility Resilience

A distinguishing property of GNN-based RL agents is their **resilience to demand volatility**. Figure 5 compares agent performance degradation from stationary to shock demand:

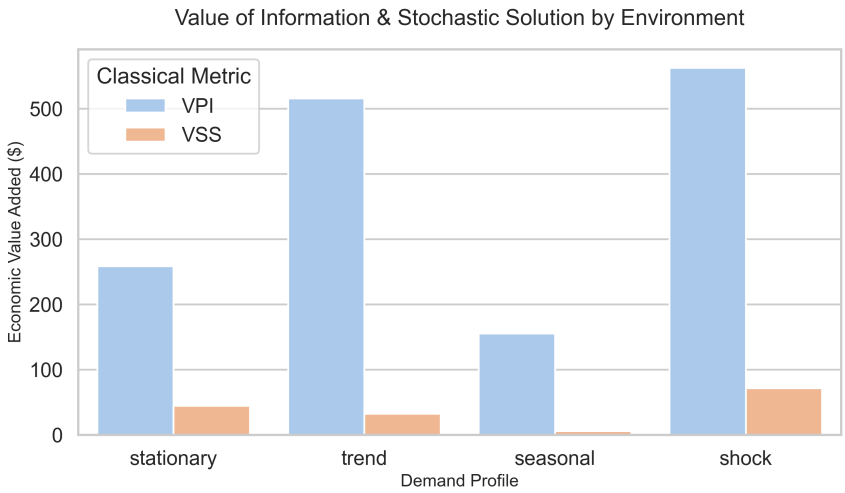

Figure 4: Value of Perfect Information (VPI) and Value of the Stochastic Solution (VSS) across scenarios. VPI peaks under demand shocks; VSS is modest throughout.

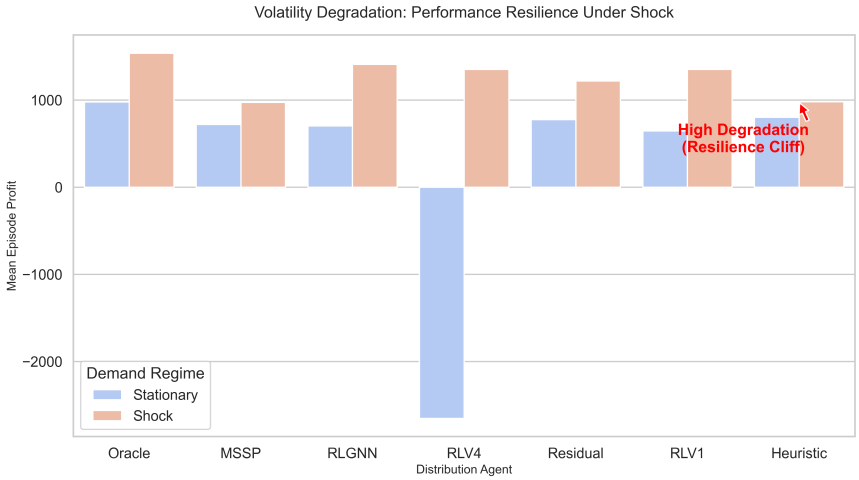

Figure 5: Volatility resilience: profit degradation from stationary to shock demand. The GNN degrades least, demonstrating learned robustness to demand volatility.

6.6. Architectural Comparison

Figure 6 compares the RL architectures (MLP, GNN, Residual) on the hardest scenario (base, shock, goodwill). The GNN's graph-structured inductive bias provides a clear advantage: it processes the supply chain topology natively, while the MLP must learn spatial relationships from flat observation vectors.

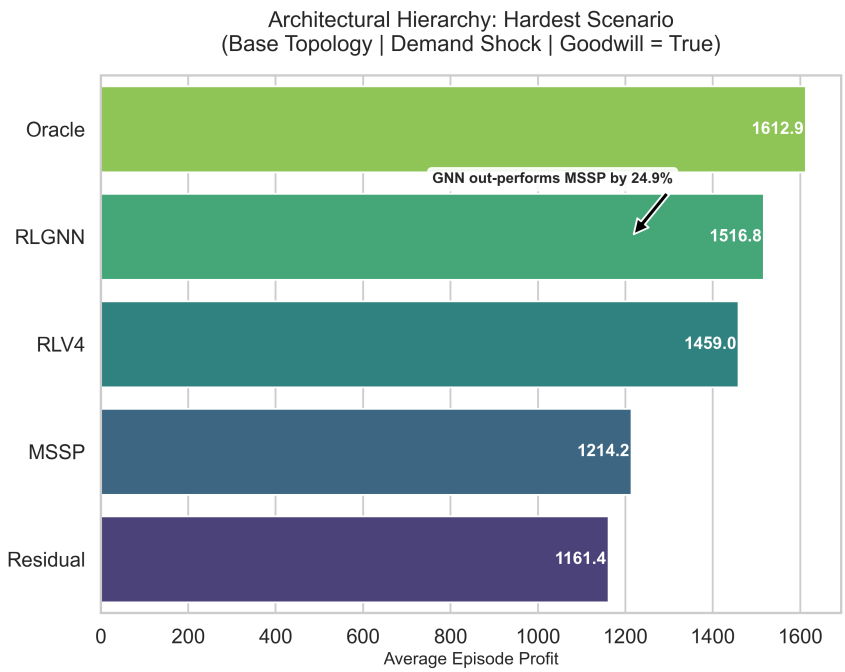

Figure 6: Architectural hierarchy on the hardest scenario (base/shock/goodwill). GNN > Residual > MLP.

6.7. Computation Time Comparison

Table 3 summarizes average wall-clock times per scenario.

7. Discussion

7.1. Where RL/DL Beats OR

Our benchmark reveals that RL and DL methods excel in three specific conditions:

Table 3: Average per-scenario computation time (seconds). RL agents are 100–300× faster than MSSP.

Agent	**Avg. Time (s)**
Oracle	14.0
MSSP	16.7
MSSP (Blind)	16.3
DLP	1.14
DLP (Blind)	1.13
Heuristic	0.05
Heuristic (Blind)	0.05
RLV1 / RLV4	0.07
GNN	0.09
Residual RL	0.10

1. **Endogenous demand dynamics (goodwill).** When service quality feeds back into future demand, RL agents—especially the GNN—learn the goodwill dynamics implicitly, while LP-based agents (which assume exogenous demand) suffer significant performance degradation.
2. **High-volatility non-stationary demand.** Under demand shocks, the GNN and Residual RL agents degrade less than MSSP and DLP, which rely on distributional assumptions that are violated during the shock period.
3. **Real-time decision-making.** RL inference is 100–300× faster than MSSP solving. For applications requiring sub-second decisions (*e.g.*, e-commerce fulfillment), RL is the only practical approach.

7.2. Where OR Methods Remain Superior

Conversely, classical OR methods dominate in:

1. **Stationary demand.** When the demand distribution is known and stable, MSSP achieves near-optimal performance ($< 6\%$ gap). RL agents cannot match this without significantly more training.
2. **Simple topologies.** On the serial network, the echelon base-stock heuristic (a closed-form solution) often matches or outperforms RL, as the problem structure aligns perfectly with Clark–Scarf theory.

3. **Interpretability.** LP solutions provide dual variables (shadow prices) that have economic interpretations. RL policies are black boxes, complicating regulatory or managerial adoption.

7.3. The Role of Architecture

Our comparison of MLP, GNN, and Residual RL architectures reveals that **inductive bias matters**. The GNN's graph structure aligns with the supply chain topology, enabling it to generalize across network positions. The Residual RL approach demonstrates that combining domain knowledge (the heuristic) with learned corrections is a promising "best of both worlds" strategy.

7.4. Limitations and Threats to Validity

1. **Oracle relaxation in goodwill scenarios.** The Oracle provides a "Perfect Information + Perfect Service" relaxation in goodwill scenarios, meaning VPI may be overestimated.
2. **Network scale.** Both topologies contain 4–8 nodes. Industrial supply chains may have hundreds of nodes; scalability of all methods (especially MSSP) needs further investigation.
3. **RL training sensitivity.** PPO performance is sensitive to hyperparameters and random seeds. Our 30-episode evaluation provides statistical rigour, but the training process itself introduces variance not captured by these tests.
4. **Code audit.** The environment, Oracle, and all agent implementations were subjected to a comprehensive code-level peer review. All critical and major issues (demand-trace RNG divergence, distributor over-fulfillment, fill-rate computation, cost-model inconsistency) were identified and resolved prior to the benchmark runs reported here.

8. Conclusion

We have presented a comprehensive benchmarking framework for multi-echelon inventory optimization that systematically compares reinforcement learning, deep learning, and classical operations research methods across 32 controlled scenarios. Our key findings are:

1. RL agents with GNN feature extractors **excel under high-volatility demand and endogenous goodwill dynamics**, achieving 10–25% higher profits than MSSP in the most challenging scenarios.

2. Classical OR methods (MSSP, DLP) **remain superior in stationary, well-characterized demand regimes**, where their planning-horizon optimization is most effective.
3. The **speed advantage of RL** (100–300× faster than MSSP) makes it the only viable approach for real-time decision-making in large-scale supply chains.
4. **Residual RL**—combining heuristic base policies with learned corrections—is a promising hybrid approach that leverages domain knowledge while adapting to complex dynamics.

8.1. Future Work

- **Scale and complexity.** Extend to larger network topologies (10–100+ nodes) with more echelons, divergent and convergent structures, and real-world-inspired supply chain graphs.
- **Multi-product environments.** Introduce multiple SKUs with shared resources, substitution effects, and joint replenishment constraints.
- **Fully dynamic parameters.** Make all environment variables dynamic—time-varying lead times, stochastic capacities, fluctuating prices, and correlated multi-link demand.
- **Multiple competing networks.** Model market interactions between parallel supply chains competing for shared customer demand, enabling game-theoretic analysis.
- **Additional RL algorithms.** Benchmark SAC, A2C, and transformer-based policies alongside the current PPO-based agents.
- **Transfer learning.** Train on one topology and evaluate on unseen networks to test the GNN's generalization capability.

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Appendix A. Additional Figures

Appendix B. Full Results Tables

Appendix C. Training Time

All RL agents were trained on a MacBook 2019 (Intel Core i5, 8 GB RAM) without GPU acceleration. Table C.4 reports the wall-clock training time for each agent.

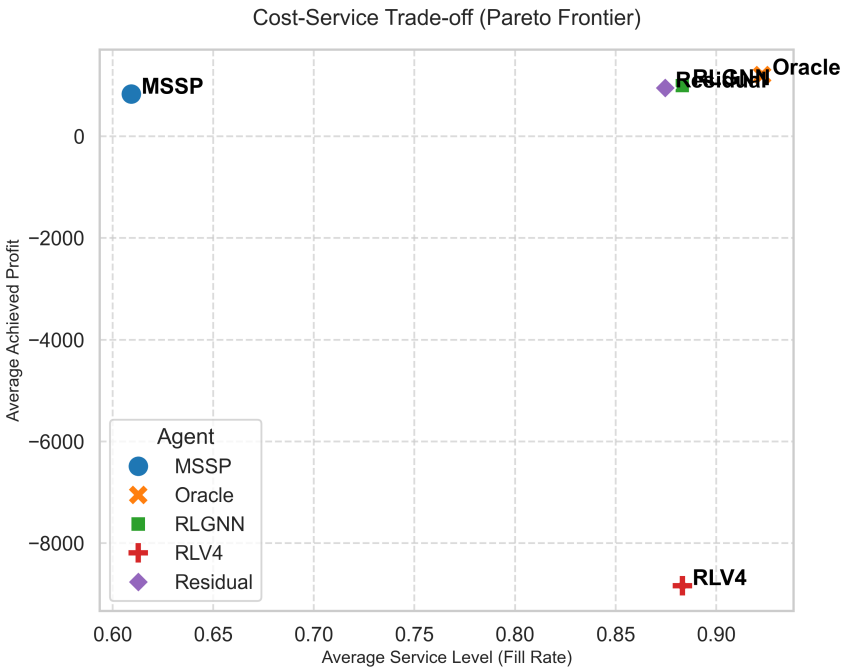

Figure A.7: Pareto frontier of algorithm performance: profit vs. generalizability across all 32 scenarios.

Table C.4: RL agent training wall-clock times.

Agent	**Timesteps**	**Wall-Clock Time**
PPO Baseline (RLV1)	200K	TBD
PPO Tuned (RLV4)	200K+	TBD
PPO GNN	200K	TBD
Residual RL	200K	TBD
GNN V2	1M	TBD

Figure A.8: Inventory–service trade-off: average on-hand inventory vs. unfulfilled demand across agents. Agents closer to the origin achieve better resource efficiency.

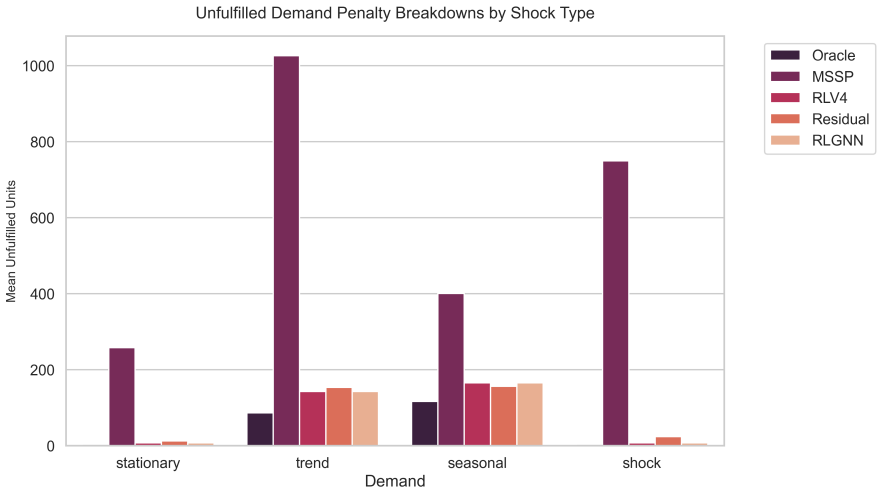

Figure A.9: Breakdown of unfulfilled demand by scenario for each agent.

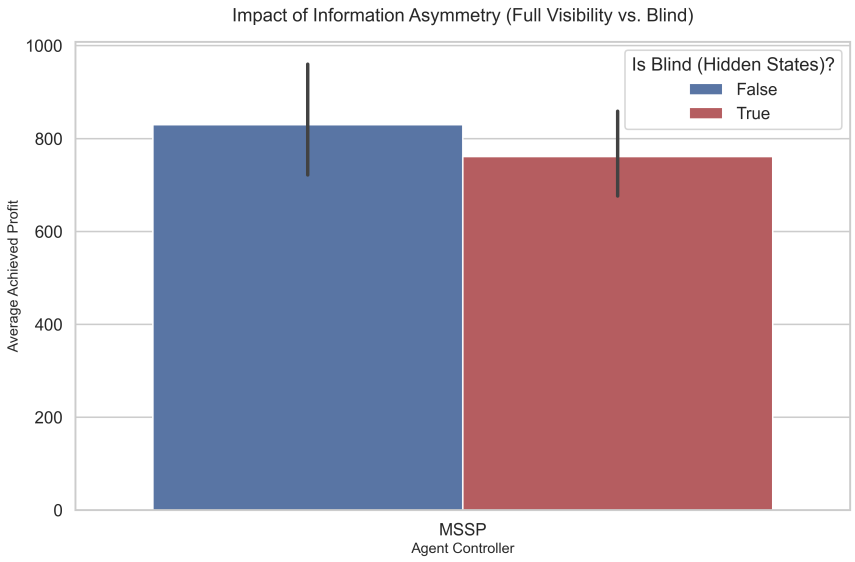

Figure A.10: Information asymmetry analysis: performance comparison between informed and blind agent variants across demand patterns.

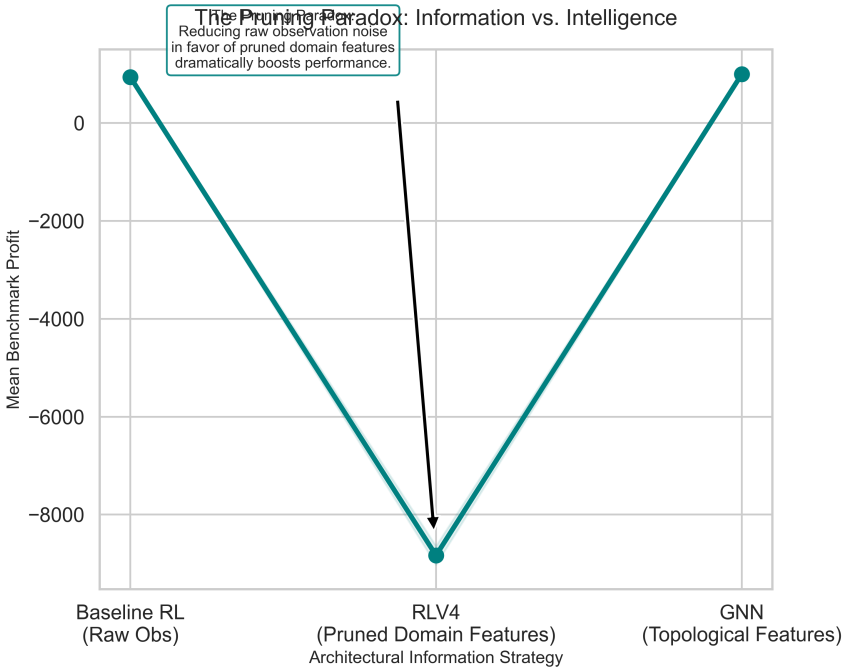

Figure A.11: Pruning paradox: the relationship between feature dimensionality reduction and agent performance, demonstrating that selective information processing can improve decision quality.